EXPERIMENT NO. 02

Aim:

Convert the ER diagram into a relational schema. Write complete MYSQL CREATE TABLE statements with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL constraints. Insert sample data demonstrate referential integrity violations.

```
DROP DATABASE IF EXISTS ecommerce;
CREATE DATABASE ecommerce;
USE ecommerce;

-- Customer
CREATE TABLE customer(
        customerid INT PRIMARY KEY AUTO_INCREMENT,
        customername VARCHAR(100) NOT NULL,
        email VARCHAR(100) NOT NULL UNIQUE
);
-- customerphone multivalued.
CREATE TABLE customer_phone(
        customerid INT NOT NULL,
        phone VARCHAR(20) NOT NULL,
        PRIMARY KEY (customerid, phone),
        FOREIGN KEY (customerid) REFERENCES customer
        (customerid)
         ON DELETE CASCADE
);
```

```
-- address
CREATE TABLE address(
        customerid    INT    NOT NULL ,
        addressid     INT    NOT NULL,
        street        VARCHAR(100)  NOT NULL,
        city          VARCHAR(50)   NOT NULL,
        state         VARCHAR(50)   NOT NULL,
        pincode       VARCHAR(10)   NOT NULL,
        PRIMARY KEY (customerid , addressid) ,
        FOREIGN KEY (customerid) REFERENCES Customer (customerid)
        ON DELETE CASCADE );

-- seller
CREATE TABLE seller (
        sellerid     INT PRIMARY KEY AUTO_INCREMENT,
        sellername   VARCHAR(100) NOT NULL UNIQUE
);
CREATE TABLE seller_phone (
        sellerid    INT   NOT NULL,
        phone       VARCHAR(20) NOT NULL,
        PRIMARY KEY (sellerid , phone),
        FOREIGN KEY (sellerid) REFERENCES seller (sellerid)
         ON DELETE CASCADE
);
```

```
-- category

CREATE TABLE category (
    categoryid INT PRIMARY KEY AUTO_INCREMENT,
    categoryname VARCHAR(100) NOT NULL UNIQUE
);

-- Product

CREATE TABLE product (
    productid INT PRIMARY KEY AUTO_INCREMENT,
    productname VARCHAR(100) NOT NULL,
    price DECIMAL(10,2) NOT NULL,
    brand VARCHAR(100),
    sellerid INT NOT NULL,
    categoryid INT NOT NULL,
    FOREIGN KEY (sellerid) References seller(sellerid)
        ON DELETE CASCADE,
    FOREIGN KEY (categoryid) References category(categoryid)
        ON DELETE CASCADE SET NULL);

-- payment

    CREATE TABLE payment (
        paymentid INT PRIMARY KEY NOT NULL AUTO_INCREMENT,
        paymentmode VARCHAR(50) NOT NULL,
        amount DECIMAL(10,2) NOT NULL,
        paymentdate DATE NOT NULL
    );
```

```
-- delivery

CREATE TABLE delivery(
    deliveryid    INT    PRIMARY KEY AUTO_INCREMENT,
    Status        VARCHAR(50)  NOT NULL,
    deliverydate  DATE
);

--orders.

CREATE  TABLE  orders (
    orderid     INT   PRIMARY KEY  AUTO_INCREMENT,
    orderdate   DATE  NOT NULL,
    totalamount DECIMAL (10,2)  NOT NULL,
    customerid  INT   NOT NULL,
    paymentid   INT   NOT  NULL  UNIQUE,
    deliveryid  INT   NOT  NULL  UNIQUE,
    FOREIGN KEY (customerid)  REFERENCES  customer (customerid)
       ON DELETE CASCADE,
    FOREIGN KEY (paymentid) REFERENCES  payment (paymentid)
       ON DELETE SET NULL,
    FOREIGN  KEY (deliveryid) REFERENCES  delivery (deliveryid)
       ON  DELETE SET NULL) ;
```

-- order item

```
CREATE TABLE orderitem(
    orderid  INT  NOT NULL,
    itemno   INT  NOT NULL,
    quantity INT  NOT NULL,
    unit_price DECIMAL(10,2) NOT NULL,
    productid  INT  NOT NULL,
    PRIMARY KEY (orderid, itemno),
    FOREIGN KEY (orderid) REFERENCES orders(orderid)
      ON DELETE CASCADE
    FOREIGN KEY (productid) REFERENCES product(productid)
      ON DELETE RESTRICT
);

-- Sample data.

INSERT INTO customer (customerid, customername, email) values
  (1, 'Ravi Kumar', 'ravi.kumar@gmail.com'),
  (2, 'Anita Sharma', 'anita.sharma@gmail.com'),
  (3, 'Mohit Verma', 'mohit.verma@gmail.com');

INSERT INTO customer_phone (customerid, phone) values
(1,'9876700009'), (1,'91234699 0')
(2,'867897821'),
(3,'726977959');
```

```
INSERT INTO address (customerid, addressid, street, city, state, pincode)
VALUES (1, 1, '12 MG Road', 'Lucknow', 'Uttar Pradesh', '226001'),
(2, 1, '45 Park Street', 'Kolkata', 'West Bengal', '700016'),
(3, 1, '78 Brigade Road', 'Bangalore', 'Karnataka', '560025');

INSERT INTO seller (sellerid, sellername) values
(1, 'TechWorld Store'),
(2, 'Fashion Hub'),
(3, 'HomeEssential');

INSERT INTO seller-phone (sellerid, phone) values
(1, '9999900000'), (2, '8800997798'), (3, '5555544444');

INSERT INTO category (categoryid, categoryname) VALUES
(1, 'Electronics'), (2, 'Clothing'), (3, 'Home & Kitchen');

INSERT INTO product (productid, productname, price, brand, sellerid,
categoryid) VALUES
(1, 'Wireless Mouse', 599.00, 'Logitech', 1, 1),
(2, 'Bluetooth Headphones', 1999.00, 'Boat', 1, 1),
(3, 'Men's Cotton T-shirt', 499.00, 'Levis', 2, 2),
(4, 'Non-Stick Pan', 899.00, 'Prestige', 3, 3);
```

```
INSERT INTO payment (paymentid, paymentmode, amount, paymentdate)
VALUES
(1, 'Credit Card', 2598.00, '2026-08-10'),
(2, 'UPI', 499.00, '2026-08-15'),
(3, 'Cash on Delivery', 899.00, '2026-08-18');

INSERT INTO delivery (deliveryid, status, deliverydate) values
(1, 'Delivered', '2026-08-14'),
(2, 'In Transit', '2026-08-20'),
(3, 'Pending', NULL);

INSERT INTO orders (orderid, orderdate, totalamount, customerid,
(1, '2026-08-10', 2598.00, 1, 1, 1),
(2,'2026-08-15', 499.00, 2, 2, 2),
(3, '2026-08-16', 899.00, 3, 3, 3);

INSERT INTO order_item (orderid, itemno, quantity, unit_price, productid)
values
(1, 1, 1, 599.00, 1)
(1, 2, 1, 1999.00, 2),
(2, 1, 1, 499.00, 3),
(3, 1, 1, 899.00, 4);
```

```
INSERT INTO order_item (orderid, itemno, quantity, unit_price, productid)
values (1, 3, 1, 250.00, 999);

/*
Error code : 1452 . Cannot add or update a child row : a
foreign key constraint fails ('ecommerce'. 'order_item', CONSTRAINT,
'Order_item_ibfk_2' FOREIGN KEY ('productid') REFERENCES
'product' ('productid') ON DELETE RESTRICT )
*/

INSERT INTO orders (orderdate, totalamount, customerid, paymentid,
deliveryid) Values (4, '2026-08-21', 1000.00, 999, NULL, NULL);

/*
Error code : 1452 : Cannot add or update a child row : a
foreign key constraint fails ('ecommerce', 'orders', CONSTRAINT,
'orders_ibfk_1' FOREIGN KEY ('customerid') REFERENCES 'customer'
('customerid') ON DELETE CASCADE )
*/

DELETE FROM product where productid = 1;

/*
Error code : 1451 . Cannot delete or update a parent row : a
foreign key constraint fails ('ecommerce'. 'order_item', CONSTRAINT,
'order_item_ibfk_2' FOREIGN KEY ('Productid') REFERENCES 'product'
('productid') ON DELETE RESTRICT )
*/
```

```
SELECT * FROM customer_phone WHERE customerid = 3;
/*
output:
```

	customerid	phone
▶	3	9012345678
*	NULL	NULL

```
*/
SELECT * FROM address WHERE customerid = 3;
/*
output:
```

	customerid	addressid	street	city	state	pincode
	3	1	78 Brigade Road	Bangalore	Karnataka	560025
	NULL	NULL	NULL	NULL	NULL	NULL

```
*/
SELECT * FROM orders WHERE customerid = 3;
/*
output:
```

	orderid	orderdate	totalamount	customerid	Paymentid	deliveryid
	3	2026-08-18	899.00	3	3	3

```
*/
SELECT * FROM order_item where orderid = 3;
/*
Error code: 1451. Cannot delete or update a parent row:
a foreign key constraint fails ('ecommerce', 'order_item', CONSTRAINT,
'order_item_ibfk_2' FOREIGN KEY ('productid') REFERENCES 'product'
('productid') ON DELETE RESTRICT)
*/
```

```
SELECT productid , productname, categoryid FROM product WHERE
categoryid = 3;

/*

output :
```

	productid	productname	categoryid
	4	Non-stick Pan	3

```
*/
DELETE FROM category WHERE categoryid = 3 ;

SELECT productid, productname, categoryid FROM product
WHERE productid = 4;

/*

output :
```

	productid	productname	categoryid
	4	Non-stick pan	NULL

```
*/

SELECT orderid , paymentid from orders where orderid = 2;

/*
```

	orderid	paymentid
	2	2

```
*/
```

```
DELETE FROM payment WHERE paymentid = 2;

SELECT orderid, paymentid from orders where orderid = 2;

/*
```

	orderid	paymentid
	2	NULL

```
*/
```

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